TGDBEK

Tensor Greedy Double Block Extended Kaczmarz for Inconsistent Tensor Linear Systems

Jeremie Mabiala

May 13, 2026

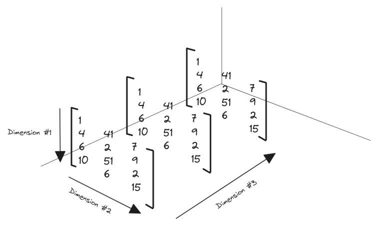
AIMS / AMMI, Senegal

What problem are we solving?

We want to solve a **large, noisy tensor system**:

$$\mathcal{A} * \mathcal{X} = \mathcal{B} + \varepsilon$$

- $\mathcal{A}, \mathcal{B}, \mathcal{X}$ are **third-order tensors**
- $*$ is the **t-product** (Kilmer et al., 2011)
- The system is **inconsistent** because of noise ε
- Target: the **minimum-norm least-squares** solution $\mathcal{X}_* = \mathcal{A}^\dagger * \mathcal{B}$



Applications: color image deblurring, MRI reconstruction, multi-channel data

The t-product: matrix algebra lifted to tensors

A third-order tensor $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$ has frontal slices $A_1, \ldots, A_{N_3}$.

The **t-product** $\mathcal{A} * \mathcal{B}$ is defined via the *block-circulant matrix*:

$$\mathrm{bcirc}(\mathcal{A}) = \begin{pmatrix} A_1 & A_{N_3} & \cdots & A_2 \\ A_2 & A_1 & \cdots & A_3 \\ \vdots & & \ddots & \vdots \\ A_{N_3} & \cdots & \cdots & A_1 \end{pmatrix}$$

Why this matters

The t-product gives tensors the *same algebra as matrices*:

- Transpose, inverse, pseudoinverse $\mathcal{A}^\dagger$
- Frobenius & spectral norms
- Orthogonal projectors

$\Rightarrow$ We can run **Kaczmarz-type iterations** directly on the tensor system.

Existing methods — and the gap

Method	Handles inconsistency?	Needs fixed partitions?
TREK (Huang et al., 2023)	**Yes**	**No**
TREBK (Huang et al., 2023)	**Yes**	**Yes**
TREGBK (Huang et al., 2023)	**Yes**	**Yes**
TREABK (An et al., 2024)	**Yes**	**Yes**
TGDBEK (ours)	**Yes**	**No**

The problem with fixed partitions

TREBK, TREGBK, TREABK all require the user to *pre-split* rows and columns of the system into blocks before running. This is **problem-dependent**, and performs poorly on **heterogeneous or sparse tensors** where the natural block structure is unknown.

Our idea: let the residual choose the blocks

Key insight. At each iteration, instead of using fixed partitions, we *greedily* pick the rows and columns with the **largest residual**.

Z-step — push $\mathcal{Z}$ to null$(\mathcal{A}^\top)$

Select column blocks U_k where

$$\frac{\|\mathcal{A}^\top_{:,j,:} * \mathcal{Z}^{(k)}\|_F^2}{\|\mathcal{A}_{:,j,:}\|_F^2} \geq \eta \cdot \max_j(\cdots)$$

Update:

$$\mathcal{Z}^{(k+1)} = \mathcal{Z}^{(k)} - \mathcal{A}_{:,U_k,:} * (\mathcal{A}_{:,U_k,:})^\dagger * \mathcal{Z}^{(k)}$$

X-step — push $\mathcal{X}$ to least-squares solution

Select row blocks J_k where residual is largest, then update:

$$\mathcal{X}^{(k+1)} = \mathcal{X}^{(k)} + (\mathcal{A}_{J_k,:,:})^\dagger * (\mathcal{B}_{J_k} - \mathcal{Z}^{(k+1)}_{J_k} - \mathcal{A}_{J_k} * \mathcal{X}^{(k)})$$

- $\eta \in (0, 1]$ — threshold controlling how many blocks are selected
- **No partition needed**: the active sets U_k, J_k are built fresh each step

What we prove

Theorem (linear convergence)

The iterates $\{\mathcal{X}^{(k)}\}$ of TGDBEK satisfy

$$\|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2 \;\le\; \alpha^k \|\mathcal{X}^{(0)} - \mathcal{X}_*\|_F^2 \;+\; C \cdot \beta^k$$

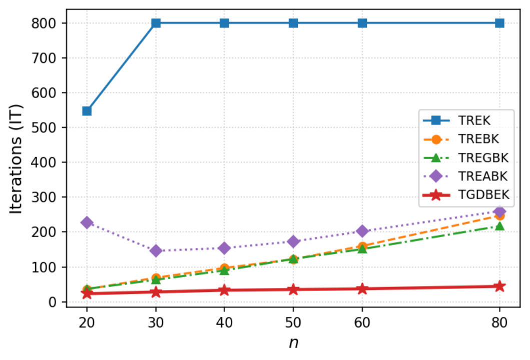
for constants $\alpha, \beta \in (0,1)$ depending on η and the singular values of $\mathcal{A}$. The iterates converge **linearly** to the minimum-norm least-squares solution $\mathcal{X}_* = \mathcal{A}^\dagger * \mathcal{B}$.

How the proof works:

1. The Z-step is an orthogonal projection $\Rightarrow$ contracts $\|\mathcal{Z}^{(k)} - \mathcal{B}^\perp\|_F$ by factor $\beta < 1$ per step.
2. The greedy row selection ensures the X-step always makes progress $\Rightarrow$ contracts $\|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F$ by $\alpha < 1$.
3. Combining the two gives the coupled linear recurrence above.

Result 1 — Dense overdetermined systems

$\mathcal{A} \in \mathbb{R}^{500\times n\times 10}$, varying n, noise $a = 10^{-3}$



Method	IT (n=80)	Converges?
TREK	800 (budget)	No
TREBK	247	Yes
TREGBK	217	Yes
TGDBEK	**44**	Yes

- **Up to 16× fewer iterations** than TREK
- **3–5× fewer** than TREBK / TREGBK
- TREK *never converges* for $n \geq 30$

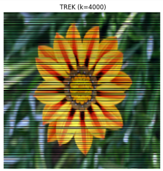
Result 2 — Color image deblurring

$200 \times 200 \times 3$ image, Gaussian blur $\sigma = 4$, noise $a = 10^{-2}$, max 1000 iterations

True

Blurred+noisy

TGDBEK



TREK (stagnates)

TREGBK (stagnates)

Method	IT	RSE
TREK	1000	7.9×10^{-1}
TREBK	1000	2.2×10^{-2}
TREGBK	1000	1.3×10^{-3}
TGDBEK	**503**	$\mathbf{1.0 \times 10^{-5}}$

Only TGDBEK converges
within the 1000-iteration budget.

PSNR ≈ 30.3 dB, SSIM ≈ 0.843

Result 3 — Harder regime & MRI-like reconstruction

High-iteration image deblurring

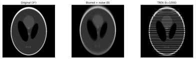
(noise $a = 10^{-1}$, max 3000 iterations)

Method	IT	CPU (s)
TREK	3000	10.7
TREBK	2363	21.7
TREGBK	1730	31.6
TGDBEK	**502**	**18.5**

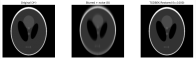
42% faster wall time than TREGBK
at one-third the iterations

Shepp-Logan MRI phantom

($128 \times 128 \times 27$, noise $a = 10^{-3}$)



TREK
18.1 dB



TGDBEK
58.0 dB

TREABK diverges (NaN). TGDBEK matches best
quality (TREGBK) with lower CPU.

Summary

What TGDBEK is:

- An iterative solver for *inconsistent* tensor linear systems
- Greedy block selection — *no user-defined partitions*
- Provably convergent to $\mathcal{X}_* = \mathcal{A}^\dagger * \mathcal{B}$

Main improvements over prior methods:

- Up to **16× fewer iterations** (vs. TREK)
- Up to **5× fewer iterations** (vs. TREBK/TREGBK)
- **Only method to converge** on color image deblurring (1000-iter budget)
- **42% wall-time reduction** on hard imaging problem

Future directions:

1. Adaptive η during iteration
2. Randomized + greedy hybrid
3. Higher-order tensors
4. Tighter convergence rate bounds

Code:
`https://github.com/jnlandu/tensor-greedy-double-extended-kaczmarz`